

Proposal: Official, versioned distribution of the Hiero SDK TCK

Background

Today the TCK can only be used by cloning `hiero-ledger/hiero-sdk-tck` and running `npm install`, or by pulling the Docker image `ivaylogarnev/hiero-tck-client` from a personal Docker Hub account.

- The TCK is not on npm: `package.json` has no name, so it cannot be published.
- SDK repositories that want to run the TCK in CI or locally have no official, versioned artifact they can pin and update.

Issue 1: Publish the TCK Docker image under the `hiero-ledger` organization

Goal: A language-neutral, official way to run the TCK, also for SDKs that do not use Node.js (Java, Go, Rust, Swift, C++).

Tasks

- Publish the image to an organization-owned registry, e.g. `ghcr.io/hiero-ledger/hiero-sdk-tck`, from a GitHub Actions release workflow (instead of the manual task `release-hiero-tck-client`).
- Tag every release with its version (e.g. `0.13.0`); `latest` points to the newest release.
- Document how to get the report out of the container (volume mount for `mochawesome-report`).
- Update the README; deprecate the image on the personal account.

Acceptance criteria

- `docker run ghcr.io/hiero-ledger/hiero-sdk-tck:<version>` runs the TCK against a JSON-RPC server given via environment variables.
- The image is built and pushed by CI for every release tag.

Issue 2: Publish the TCK as a versioned CLI package on npm

Goal: SDK repositories can pin the TCK as a dependency (e.g. `@hiero-ledger/hiero-sdk-tck`), update it with Renovate/Dependabot and run it without cloning: `npx @hiero-ledger/hiero-sdk-tck --server http://localhost:8544 --test AccountCreate`.

Tasks

- Give the package a scoped name and a `bin` entry with a small CLI: JSON-RPC server URL; network profile (`local`, `testnet`, custom via env file or variables); test selection (test names as in the Docker image, or file patterns); report directory.
- Ship compiled JavaScript instead of running from source with `ts-node` and `tsconfig-paths`.
- Read the configuration from CLI options, environment variables or an env file in the caller's working directory, never from files inside the package. Write the report to the caller's working directory (or a given directory).
- Publish from CI on release tags with npm provenance (Trusted Publishing), using the same version as the Docker image.

Acceptance criteria

- In an empty directory, `npx @hiero-ledger/hiero-sdk-tck@<version> --server <url> --test AccountCreate` runs the tests and writes the report locally.
- Every release tag publishes the npm package and the Docker image with the same version.

Note

Both issues are independent. Issue 1 matters more for non-JavaScript SDKs, Issue 2 for local development and Node-based repositories.

Namespaces

The names follow the Hiero artifact namespace guidelines ([github.com/hiero-ledger/governance, rules-and-guidelines/namespaces.md](https://github.com/hiero-ledger/governance/blob/master/rules-and-guidelines/namespaces.md)):

- npm scope `@hiero-ledger`, artifact names prefixed with `hiero-: @hiero-ledger/hiero-sdk-tck`.
- Container registry `ghcr.io/hiero-ledger`, image name matching the owning repository:
`ghcr.io/hiero-ledger/hiero-sdk-tck`.
- Publishing only by the maintainers or the release automation of the TCK repository.
- Both names must be documented in the TCK README before the first publication.